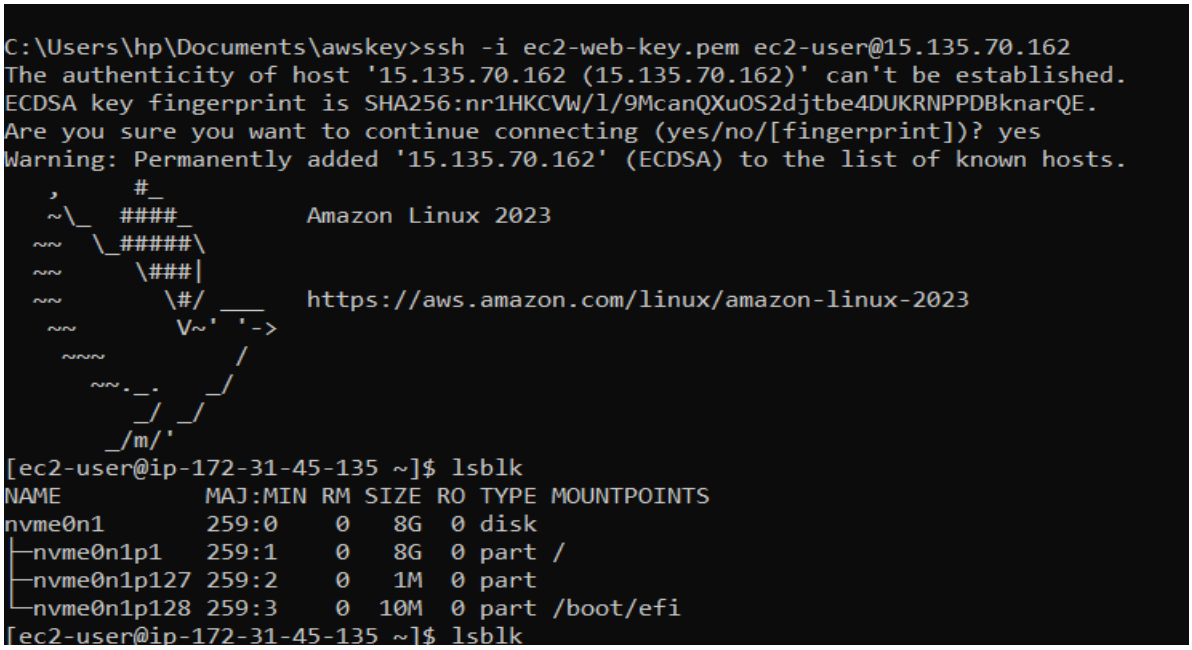
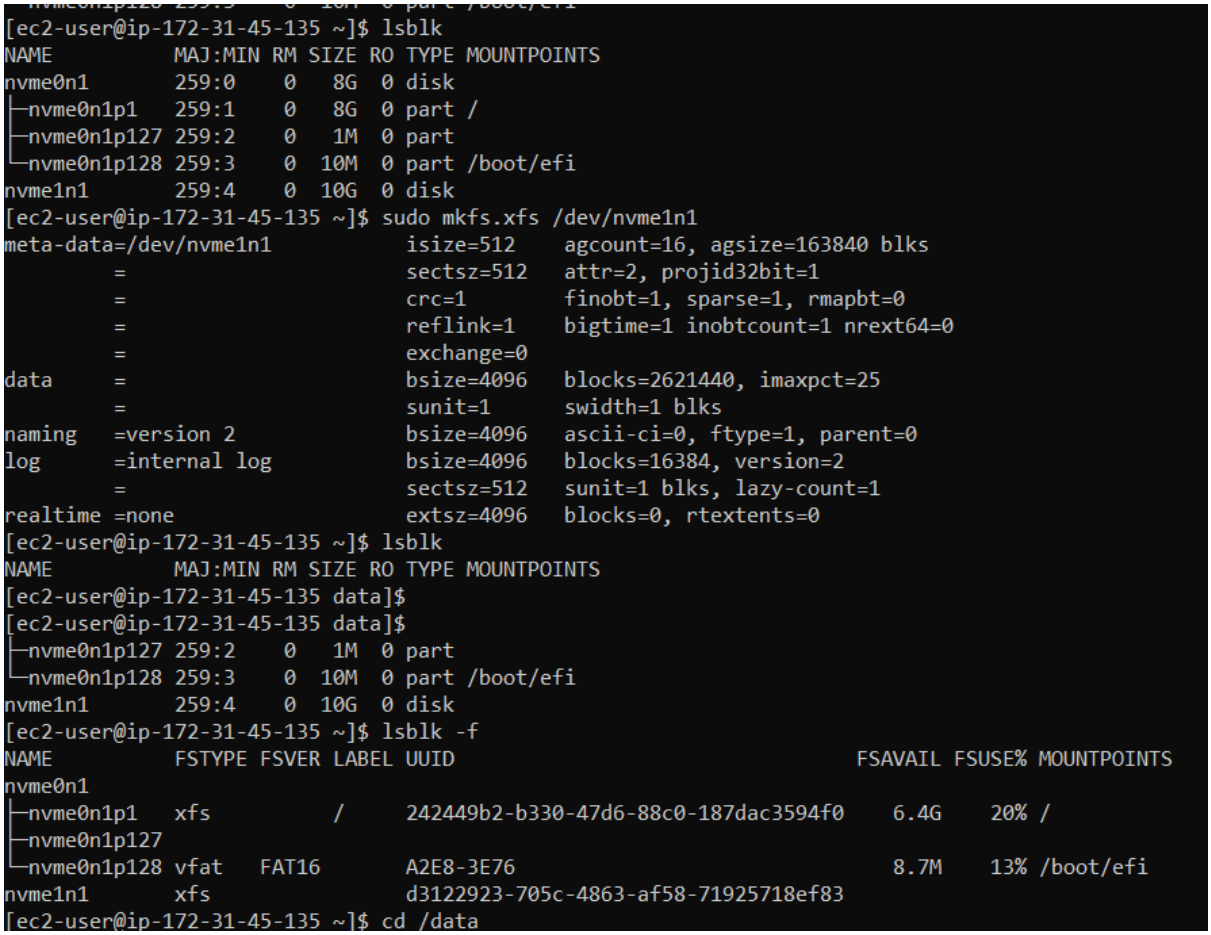
EBS_PRACTICE_SCREENSHOTS

- Successfully connected to aws instance



- create file system type for newly attached EBS.



```
nvme1n1       259:4    0  10G  0 disk
[ec2-user@ip-172-31-45-135 ~]$ lsblk -f
NAME          FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
nvme0n1
├─nvme0n1p1   xfs          /     242449b2-b330-47d6-88c0-187dac3594f0    6.4G    20% /
├─nvme0n1p127
└─nvme0n1p128 vfat   FAT16       A2E8-3E76                               8.7M    13% /boot/efi
nvme1n1       xfs                d3122923-705c-4863-af58-71925718ef83
```

- Created a dir for mount the storage
- The dir name called /data

```
mkdir: cannot create directory '/data': Permission denied
[ec2-user@ip-172-31-45-135 ~]$ sudo mkdir /data
[ec2-user@ip-172-31-45-135 ~]$ ls -l /
total 32
lrwxrwxrwx.   1 root root     7 Jan 30  2023 bin -> usr/bin
dr-xr-xr-x.   5 root root 16384 Oct 21 17:49 boot
drwxr-xr-x.   2 root root     6 Oct 23 05:30 data
drwxr-xr-x.  14 root root  3180 Oct 23 05:27 dev
drwxr-xr-x.  76 root root 16384 Oct 23 04:50 etc
drwxr-xr-x.   3 root root    22 Oct 23 04:50 home
lrwxrwxrwx.   1 root root     7 Jan 30  2023 lib -> usr/lib
lrwxrwxrwx.   1 root root     9 Jan 30  2023 lib64 -> usr/lib64
drwxr-xr-x.   2 root root     6 Oct 21 17:45 local
drwxr-xr-x.   2 root root     6 Jan 30  2023 media
drwxr-xr-x.   2 root root     6 Jan 30  2023 mnt
drwxr-xr-x.   3 root root    17 Oct 21 17:49 opt
dr-xr-xr-x. 114 root root     0 Oct 23 04:50 proc
dr-xr-x---.   3 root root   103 Oct 21 17:49 root
drwxr-xr-x.  28 root root   840 Oct 23 04:50 run
lrwxrwxrwx.   1 root root     8 Jan 30  2023 sbin -> usr/sbin
drwxr-xr-x.   2 root root     6 Jan 30  2023 srv
dr-xr-xr-x.  13 root root     0 Oct 23 04:50 sys
drwxrwxrwt.  11 root root   220 Oct 23 05:33 tmp
drwxr-xr-x.  12 root root   144 Oct 21 17:47 usr
drwxr-xr-x.  19 root root   266 Oct 23 04:50 var
```

- After that mounted temperorly

```
drwxr-xr-x.  19 root root   266 Oct 23 04:50 var
[ec2-user@ip-172-31-45-135 ~]$ sudo mount /dev/nvme1n1 /data
[ec2-user@ip-172-31-45-135 ~]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk /data
[ec2-user@ip-172-31-45-135 ~]$ df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme1n1     10G  104M  9.9G   2% /data
[ec2-user@ip-172-31-45-135 ~]$ df -h /data
```

- Then mounted permanently also
- For permanent mount first we want to edit the conf file /etc/fstab

ec2-user@ip-172-31-45-135:/data

```
#
UUID=242449b2-b330-47d6-88c0-187dac3594f0     /          xfs    defaults,noatime  1   1
UUID=A2E8-3E76        /boot/efi       vfat    defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2
UUID=d3122923-705c-4863-af58-71925718ef83 /data xfs defaults 0 0
~
~
```

```
[ec2-user@ip-172-31-45-135 data]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
[ec2-user@ip-172-31-45-135 data]$ sudo blkid
/dev/nvme0n1p1: LABEL="/" UUID="242449b2-b330-47d6-88c0-187dac3594f0" BLOCK_SIZE="4096"
/dev/nvme0n1p128: SEC_TYPE="msdos" UUID="A2E8-3E76" BLOCK_SIZE="512" TYPE="vfat" PARTLA
/dev/nvme0n1p127: PARTLABEL="BIOS Boot Partition" PARTUUID="55cf7f90-fca0-46dc-be28-1a2
/dev/nvme1n1: UUID="d3122923-705c-4863-af58-71925718ef83" BLOCK_SIZE="512" TYPE="xfs"
[ec2-user@ip-172-31-45-135 data]$ sudo vim /etc/fstab
[ec2-user@ip-172-31-45-135 data]$ sudo vim /etc/fstab
[ec2-user@ip-172-31-45-135 data]$ mount /dev/nvme1n1
mount: /data: must be superuser to use mount.
[ec2-user@ip-172-31-45-135 data]$ sudo mount /dev/nvme1n1
[ec2-user@ip-172-31-45-135 data]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk /data
```

```
[ec2-user@ip-172-31-45-135 data]$ ls
fazil  janeeb  noufal  rishi
[ec2-user@ip-172-31-45-135 data]$
```

AWS EBS Practice

Tasks Completed

- Connected to an EC2 instance using SSH
- Attached an EBS volume
- Created an XFS filesystem
- Created `/data` mount point
- Mounted the volume temporarily
- Configured permanent mounting using `/etc/fstab`

Commands Used

```
lsblk
sudo mkfs.xfs /dev/nvme1n1
sudo mkdir /data
sudo mount /dev/nvme1n1 /data
df -Th
```

Result

Successfully practised EBS volume mounting and filesystem configuration on an EC2 instance.